

Artem Semidetnov

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Summary

Software/research engineer building AI tooling for dependently typed programming languages, including MCP integration, coding agents, IDE tooling, and model fine-tuning. Background in mathematics, type systems, and formal verification.

Technical Skills

- **Programming languages & compilers:** Arend, dependent type systems, typechecking, language tooling, IDE integration, Lean, Haskell.
- **AI engineering:** MCP, coding agents, LangGraph, Koog, model fine-tuning, code completion, LLM evaluation.
- **Software engineering:** Kotlin, Java, Python, SQL, JVM ecosystem.
- **Formal methods:** theorem proving, proof assistants, formalization.
- **Spoken languages:** Russian (native), English C1 (IELTS 8.0/9), French (conversational).

Work Experience

- JetBrains, Research Intern in HoTT and Dependent Types Lab

Remote

Nov 2025 – Apr 2026

 - Designed and developed the MCP server for the Arend proof assistant, exposing typechecking and relevant IDE functionality to AI coding agents.
 - Integrated the MCP server with the Arend plugin for IntelliJ IDEA.
 - Built an experimental LLM coding agent for generating Arend code using the Koog framework.
 - Fine-tuned models for Arend code completion.
- JetBrains, Research Intern in HoTT and Dependent Types Lab

Pafos, Cyprus

June 2024 – Sept 2024

 - Contributed formalized results in algebra and homotopy type theory to the official Arend library.
 - Formalized the Eckmann-Hilton argument, Eilenberg-MacLane spaces, and homology of types.
 - Formalized group automorphisms, Schur’s Lemma, Maschke’s Lemma, and characterizations of group actions; selected contributions shipped in [Arend 1.10](#).
- IPONWEB (acquired by Criteo), Machine Learning Intern

Pafos, Cyprus

May 2023 – Sept 2023

 - Reverse-engineered the behavior of a Criteo ML system that analyzes websites and produces word2vec representations; evaluated potential applications.

Education

- MSc **Geneva University**, Mathematics

2025 – 2027
- BSc **Saint-Petersburg State University**, Mathematics

2021 – 2025
- BSc **Neapolis University Of Pafos**, Applied Computer Science JetBrains Program

2022 – 2024

AI-Assisted Mathematical Research

- Used AI tools in original mathematical research, resulting in two 2026 preprints: [Kourovka Notebook solutions](#) and [stable commutator length](#).

Publications

- The operad associated to a crossed simplicial group**

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<https://doi.org/10.48550/arXiv.2512.14498>

Dec 2025
- On the geometry of free nilpotent groups**

May 2021

Artem Semidetnov, Ruslan Magdiev

<https://doi.org/10.48550/arXiv.2106.00095> 